Mälardalen University

PROJECT IN MATHEMATICS

On Permutation Groups

by

Adrian Swande

MMA291 — Project in Mathematics

DIVISION OF MATHEMATICS AND PHYSICS
MÄLARDALEN UNIVERSITY
SE-721 23 VÄSTERÅS, SWEDEN

Mälardalen University

MMA291 — Project in Mathematics

Date of presentation:
XX June 2026

Project name:
The Title of the Thesis

Author:
Adrian Swande

Version: 1
20th May 2026

Supervisor:
Peder Thompson

Examiner:
To be determined

Comprising:
7.5 ECTS credits

Abstract

Under construction.

Acknowledgements

Under construction.

Contents

Chapter 1

Introduction

The motivation behind the research whose the results can be found largely within this work, was the desire to find an answer to a deceptively simple question: “What is the process by which methods for solving problems are found?”

The question arose within me in relation to a long fascination with the problem-solving process employed when finding methods for solving “twisty puzzles” – such as the classic Rubik’s cube. It was this initial focus which led me to study *permutation groups*, and initially formulate my thoughts and findings in terms thereof.

...

Chapter 2

Preliminaries

2.1 Groups

Definition 1 (Group)**.** In mathematics, a *group* G denotes a set of elements $\{a, b, c, \cdots\}$ and an accompanying binary operator $*$, together satisfying the following conditions:

1. *Closure*: For every ordered pair of elements in the group, the binary product thereof (as defined by the binary operator) is also an element of that group; $\forall a, b \in G\ \ (a * b \in G)$.

2. *Associativity*: The order wherein ordered pairs of elements are evaluated has no consequence on the restulting product; $\forall a, b, c \in G\ \ ((a * b) * c = a * (b * c))$.

3. *Existence of Identity*: There exists in the group an *identity* element, the omission of which from any expression containing any other element or elements has no consequence on the evaluation of that expression; $\exists e \in G\ \forall a \in G\ (e * a = a * e = a)$.

4. *Existence of Inverse*: Each element in the group has its own *inverse*, who's product therewith evalutes to the identity; $\forall a \in G\ \exists b \in G\ (a * b = e)$, where e is the identity.

Definition 2 (Commutativity)**.** Let G be a group. Then it is *commutative* (or *Abelian*) if $\forall a, b \in G\ (a * b = b * a)$.

The integers $\mathbb{Z} = \{\cdots, -2, -1, 0, 1, 2, \cdots\}$ together with *addition* constitutes an Abelian group with 0 acting as its identity (since $\forall a \in \mathbb{Z}\ \ (0 + a = a + 0 = a)$). The negative numbers exist as the inverses of the positives, and vice versa. We know addition is associative, and since $\mathbb{Z}$ is infinite in each direction, we will never find a pair of elements therein whose sum is not also an element therein. This group can also be expressed as additively *generated* by the element 1, since 1 and its inverse -1 together with $+$ can express the whole group. This generative definition is denoted $G = (\langle\{1\}\rangle, +) = (\mathbb{Z}, +)$. The group generated by 2 – to give another example – is the group of all *even* integers.

Definition 3 (Cyclicity)**.** A group which can be generated by a single element is called a *cyclic* group.

When the operator is given implicitly, a cyclic group with the generator g can be written as $\langle g \rangle$.

Definition 4 (Subgroup)**.** Let G be a group under a binary operator $*$. Then $H \subset G$ is a *subgroup* of G if H is also a group under $*$.

2.2 Permutations

Definition 5 (Permutation)**.** A *permutation* π of a set A is a bijective map $\pi : A \longrightarrow A$.

We write π^n for a permutation π to denote $\pi \circ \cdots \circ \pi$ where π is composed with itself $n \in \mathbb{N}$ times. π^{-n} denotes the composition $\pi^{-1} \circ \cdots \circ \pi^{-1}$ where the inverse π^{-1} is composed with itself $n \in \mathbb{N}$ times. The permutation π^0 always equals the identity.

Definition 6 (Permutation Group)**.** A *permutation group* on a set A is a group of permutations of A under composition.

Definition 7 (Support)**.** The *support* $\underline{\pi}$ of a permutation π of A is the set $\{a \in A \mid \pi(a) \neq a\}$.

A permutation whose support is the empty set equals the identity. We say that a permutation π of A *fixes* an element $a \in A$ if $a \notin \underline{\pi}$.

Definition 8 (Cycle)**.** A *cycle* is a permutation π such that $\forall a, b \in \underline{\pi}\ \exists\, \phi \in \langle \pi \rangle\ (\phi(a) = b)$.

Since the support of the identity is the empty set, the identity also a cycle.

Definition 9 (Subcycle)**.** The *subcycle* $\overset{a}{\pi}$ of an element $a \in A$ within a permutaion π on A is the cycle whose support $\underline{\overset{a}{\pi}}$ is the set $\{\phi(a) \mid \phi \in \langle \pi \rangle\}$ and where $\forall b \in \underline{\overset{a}{\pi}}\ \left(\overset{a}{\pi}(b) = \pi(b)\right)$.

Definition 10 (Orbit)**.** The *orbit* of an element $a \in A$ in a permutation π on A is the support of the subcycle of a in π.

Definition 11 (Transposition)**.** A *transposition* is a cycle π with $\left|\underline{\pi}\right| = 2$.

Theorem 1. *Let π be a permutation of a set A. Then there exists a family of cycles $(\pi_i)_{i \in I}$ on A with pairwise disjoint supports such that for each $a \in \underline{\pi}$, there exists unique $i \in I$ where $\pi(a) = \pi_i(a)$.*

Proof. Let $a_{k+1} \in \underline{\pi} \setminus \bigcup_{i \in \{1, \cdots, k\}} \underline{\pi_i}$ where $k \in \mathbb{N}$ and each π_i is a cycle satisfying $\forall n \in \mathbb{Z}\ \left(\pi_i^n(a_{k+1}) = \pi^n(a_{k+1})\right)$. Then

□

In *cycle notation*, Theorem 1 is used to express permutations in a convenient way. Here, each cycle in the expressed permutation is written as a parenthesized list of elements, where each element therein is mapped to the succeeding element save for the last one, which is mapped to the first. For example, the permutation $(a, b, c)(d, e)$ maps $a \mapsto b$, $b \mapsto c$ and $c \mapsto a$, as well as $d \mapsto e$ and $e \mapsto d$. Note that this permutation is the same as the permutation $(c, a, b)(e, d)$, but not the same as $(a, c, b)(d, e)$.

In this text, permutations containing symbols or numbers with multiple digits will be written with dividing commas, whilst permutations containing exclusively the numbers 0 through 9 will be written therewithout – for example $(1234) = (1, 2, 3, 4)$.

Lemma 1. *Let α and β be permutations of A. Then α and β do not commute if and only if there is an element $c \in \underline{\alpha} \cap \underline{\beta}$ where the cycles of α and β wherein c is contained are neither equal nor each other's inverses.*

Proof. Let α and β be permutations of A and $\underline{\alpha} \cap \underline{\beta} = \emptyset$. Then $\forall b \in A \ (\alpha(b) = b \vee \beta(b) = b)$. Thus for each $b \in A$ both $(\alpha \circ \beta)(b)$ and $(\beta \circ \alpha)(b)$ evaluate to the same expression, which will be either $\alpha(b)$, $\beta(b)$ or b. A permutation always commutes with itself, so if $\alpha_c = \beta_c$ for some $c \in \underline{\alpha} \cap \underline{\beta}$ – where α_c and β_c are the cycles of α and β wherein c occurs – then $\alpha_c \circ \beta_c = \beta_c \circ \alpha_c$. If each cycle in α and β commute, so do they. If $\alpha_c = \beta_c^{-1}$, then the cycles neutralize each other $\alpha_c \circ \beta_c = \beta_c \circ \alpha_c = e$. Now suppose there is an element $c \in \underline{\alpha} \cap \underline{\beta}$ such that $\alpha_c \neq \beta_c$ and $\alpha_c \neq \beta_c^{-1}$. Then from $\beta(c) \notin \underline{\alpha}$ follows $(\alpha \circ \beta)(c) = \beta(c)$ and thereof $(\beta \circ \alpha)(c) \neq \beta(c)$. If $\beta(c) \in \underline{\alpha}$, then $(\alpha \circ \beta)(c) \neq \beta(c)$

$\square$

Chapter 3

The Conjugates and Commutators of Permutations

Herein, all lowercase geek letters are implied to be elements in a permutation group G on A. Elements of A are represented by lowercase latin characters. Further, the operator $\circ$ will henceforth be implicit, and thus omitted from expressions. As an example, take $\pi \circ \phi = \pi\phi$.

Definition 12 (Conjugate)**.** The *conjugate* $\searrow^{\alpha}_{\beta}$ of β by α is the composition $\alpha^{-1}\beta\alpha$.

Definition 13 (Commutator)**.** The *commutator* $\nwarrow\!\!\!\!\searrow^{\alpha}_{\beta}$ of α and β is the composition $\beta^{-1}\alpha^{-1}\beta\alpha$.

These two constructions – though they can seem arbitrary at first – have interesting and useful properties when working within groups – specifically when the supports of their two permutations share elements.

In the trivial case where they share no element, the permutations commute, so the conjugate of α and β is β, and the commutator thereof is the identity; $\alpha \cap \beta = \emptyset$ implies that $\alpha^{-1}\beta\alpha = \beta$ and $\beta^{-1}\alpha^{-1}\beta\alpha = e$.

Lemma 2. *Let* $\underline{\alpha} \cap \underline{\beta} = \{c\}$*. Then* $\overset{c}{\searrow}{}^{\alpha}_{\beta} = \left(\alpha^{-1}(c), \beta(c), \cdots, \beta^{-1}(c)\right)$.

Proof. For each $b \in \underline{\alpha} \setminus \left\{\alpha^{-1}(c)\right\}$, $\alpha(b) \notin \underline{\beta}$. Since $\beta\alpha(b) = \alpha(b)$, the conjugate can be rewritten for b: $\left(\alpha^{-1}\beta\alpha\right)(b) = \left(\alpha^{-1}\alpha\right)(b) = b$. In the case of $\alpha^{-1}(c)$, $\left(\alpha^{-1}\beta\alpha\alpha^{-1}\right)(c) = \left(\alpha^{-1}\beta\right)(c) = \beta(c)$, since $\beta(c) \notin \underline{\alpha} \implies \left(\alpha^{-1}\beta\right)(c) = \beta(c)$. For each $d \in \underline{\beta} \setminus \left\{\beta^{-1}(c), c\right\}$, $d, \beta(d) \notin \underline{\alpha}$, so $\left(\alpha^{-1}\beta\alpha\right)(d) = \beta(d)$. Lastly, $\beta^{-1}(c)$ maps to $\left(\alpha^{-1}\beta\alpha\beta^{-1}\right)(c) = \left(\alpha^{-1}\beta\beta^{-1}\right)(c) = \alpha^{-1}(c)$. This gives the function

$$\searrow^{\alpha}_{\beta}(x) = \begin{cases} x & : x \in \underline{\alpha} \setminus \left\{\alpha^{-1}(c)\right\} \\ \beta(c) & : x = \alpha^{-1}(c) \\ \beta(x) & : x \in \underline{\beta} \setminus \left\{\beta^{-1}(c), c\right\} \\ \alpha^{-1}(c) & : x = \beta^{-1}(c) \end{cases} \tag{3.1}$$

whose subcycle of c is written as $\left(\alpha^{-1}(c), \beta(c), \cdots, \beta^{-1}(c)\right)$. □

Corollary 1. *If* β *is a transposition, then so is* $\searrow^{\alpha}_{\beta}$.

Proof. If β is a transposition, then $\beta = (c, \beta(c))$ and $\beta(c) = \beta^{-1}(c)$. Using (3.1) we get

$$\searrow^{\alpha}_{\beta}(x) = \begin{cases} x & : x \in \underline{\alpha} \setminus \left\{\alpha^{-1}(c)\right\} \\ \beta(c) & : x = \alpha^{-1}(c) \\ \beta(x) & : x \in \emptyset \\ \alpha^{-1}(c) & : x = \beta^{-1}(c) \end{cases} \tag{3.2}$$

$$= \begin{cases} x & : x \in \underline{\alpha} \setminus \left\{\alpha^{-1}(c)\right\} \\ \beta(c) & : x = \alpha^{-1}(c) \\ \alpha^{-1}(c) & : x = \beta(c) \end{cases} \tag{3.3}$$

$$= \left(\alpha^{-1}(c), \beta(c)\right). \tag{3.4}$$

□

Lemma 3. *Let* $\underline{\alpha} \cap \underline{\beta} = \{c\}$. *Then* $\nearrow\!\!\!\!\searrow^{\alpha}_{\beta} = \left(c, \beta^{-1}(c), \alpha^{-1}(c)\right)$.

Proof. Using (3.1) from Lemma 2, we get

$$\nearrow\!\!\!\!\searrow^{\alpha}_{\beta}(x) = \left(\beta^{-1} \searrow^{\alpha}_{\beta}\right)(x) = \begin{cases} \left(\beta^{-1}\right)(x) & : x \in \underline{\alpha} \setminus \left\{\alpha^{-1}(c)\right\} \\ \left(\beta^{-1}\beta\right)(c) & : x = \alpha^{-1}(c) \\ \left(\beta^{-1}\beta\right)(x) & : x \in \underline{\beta} \setminus \left\{\beta^{-1}(c), c\right\} \\ \left(\beta^{-1}\alpha^{-1}\right)(c) & : x = \beta^{-1}(c) \end{cases} \tag{3.5}$$

$$= \begin{cases} x & : x \in \underline{\alpha} \setminus \left\{\alpha^{-1}(c), c\right\} \\ \beta^{-1}(x) & : x = c \\ c & : x = \alpha^{-1}(c) \\ x & : x \in \underline{\beta} \setminus \left\{\beta^{-1}(c), c\right\} \\ \alpha^{-1}(c) & : x = \beta^{-1}(c) \end{cases} \tag{3.6}$$

$$= \begin{cases} x & : x \in (\underline{\alpha} \cup \underline{\beta}) \setminus \left\{\alpha^{-1}(c), \beta^{-1}(c), c\right\} \\ c & : x = \alpha^{-1}(c) \\ \alpha^{-1}(c) & : x = \beta^{-1}(c) \\ \beta^{-1}(c) & : x = c \end{cases} \tag{3.7}$$

$$= \left(c, \beta^{-1}(c), \alpha^{-1}(c)\right). \tag{3.8}$$

□

Lemma 4. *Let* $\underline{\alpha} \cap \underline{\beta} = \{c_1, \cdots, c_n\}$, $\alpha \neq \beta$ *for* $n \geq 2$ *where* $\alpha^{k-1}(c_1) = \beta^{k-1}(c_1) = c_k$ *for* $k \in \{1, \cdots, n\}$. *Then* $\overset{c}{\searrow}{}^{\alpha}_{\beta} = \left(\alpha^{-1}(c_1), c_1, \cdots, c_{n-1}, \beta(c_n), \cdots, \beta^{-1}(c_1)\right)$.

Proof. Let $a \in \left\{c_n, \alpha(c_n), \cdots, \alpha^{-2}(c_1)\right\}$. Then $\alpha(a) \notin \underline{\beta}$, so $\left(\alpha^{-1}\beta\alpha\right)(a) = \left(\alpha^{-1}\alpha\right)(a) = a$. Conversely, each $b \in \left\{\beta(c_n), \cdots, \beta^{-2}(c_1)\right\}$ maps to $\left(\alpha^{-1}\beta\alpha\right)(b) = \beta(b)$, since $b, \beta(b) \notin \underline{\alpha}$. As for the elements $\{c_1, \cdots, c_{n-2}\}$, none thereof diverge from the common path $\alpha^{k-1}(c_1) = \beta^{k-1}(c_1) = c_k$ for $k \in \{1, \cdots, n\}$, wherefore $c_j \overset{\alpha}{\mapsto} c_{j+1} \overset{\beta}{\mapsto} c_{j+2} \overset{\alpha^{-1}}{\mapsto} c_{j+1}$ in $\searrow^{\alpha}_{\beta}$ for each $j \in \{1, \cdots, n-2\}$. Lastly, $\alpha^{-1}(c_1)$ maps to $\left(\alpha^{-1}\beta\alpha\alpha^{-1}\right)(c_1) = \left(\alpha^{-1}\beta\right)(c_1) =$

$\alpha^{-1}(c_2) = c_1$, $\beta^{-1}(c_1)$ maps to $\left(\alpha^{-1}\beta\alpha\beta^{-1}\right)(c_1) = \left(\alpha^{-1}\beta\beta^{-1}\right)(c_1) = \alpha^{-1}(c_1)$ and c_{n-1} maps to $\left(\alpha^{-1}\beta\alpha\right)(c_{n-1}) = \left(\alpha^{-1}\beta\right)(c_n) = \beta(c_n)$. This gives the function

$$\searrow^{\alpha}_{\beta}(x) = \begin{cases} x & : x \in \left\{c_n, \alpha(c_n), \cdots, \alpha^{-2}(c_1)\right\} \\ c_1 & : x = \alpha^{-1}(c_1) \\ \beta(x) & : x \in \left\{\beta(c_n), \cdots, \beta^{-2}(c_1)\right\} \cup \{c_1, \cdots, c_{n-2}\} \\ \beta(c_n) & : x = c_{n-1} \\ \alpha^{-1}(c_1) & : x = \beta^{-1}(c_1) \end{cases} \tag{3.9}$$

$$\implies \overset{c_1}{\searrow}{}^{\alpha}_{\beta} = \left(\alpha^{-1}(c_1), c_1, \cdots, c_{n-1}, \beta(c_n), \cdots, \beta^{-1}(c_1)\right). \tag{3.10}$$

□

Lemma 5. *Let α and β be permutations with the same properties as those in Lemma 4. Then* $\times^{\alpha}_{\beta} = \left(\alpha^{-1}(c_1), \beta^{-1}(c_1)\right)(c_{n-1}, c_n)$.

Proof. From (3.9) from Lemma 4 we get

$$\times^{\alpha}_{\beta}(x) = \begin{cases} \beta^{-1}(x) & : x \in \left\{c_n, \alpha(c_n), \cdots, \alpha^{-2}(c_1)\right\} \\ \beta^{-1}(c_1) & : x = \alpha^{-1}(c_1) \\ \left(\beta^{-1}\beta\right)(x) & : x \in \left\{\beta(c_n), \cdots, \beta^{-2}(c_1)\right\} \cup \{c_1, \cdots, c_{n-2}\} \\ \left(\beta^{-1}\beta\right)(c_n) & : x = c_{n-1} \\ \left(\beta^{-1}\alpha^{-1}\right)(c_1) & : x = \beta^{-1}(c_1) \end{cases} \tag{3.11}$$

$$= \begin{cases} x & : x \in \left\{\alpha(c_n), \cdots, \alpha^{-2}(c_1)\right\} \\ \beta^{-1}(c_n) & : x = c_n \\ \beta^{-1}(c_1) & : x = \alpha^{-1}(c_1) \\ x & : x \in \left\{\beta(c_n), \cdots, \beta^{-2}(c_1)\right\} \cup \{c_1, \cdots, c_{n-2}\} \\ c_n & : x = c_{n-1} \\ \alpha^{-1}(c_1) & : x = \beta^{-1}(c_1) \end{cases} \tag{3.12}$$

$$= \begin{cases} x & : x \in \left\{c_{n-1}, c_n, \alpha^{-1}(c_1), \beta^{-1}(c_1)\right\} \\ \beta^{-1}(c_1) & : x = \alpha^{-1}(c_1) \\ \alpha^{-1}(c_1) & : x = \beta^{-1}(c_1) \\ c_n & : x = c_{n-1} \\ c_{n-1} & : x = c_n \end{cases} \tag{3.13}$$

$$= \left(\alpha^{-1}(c_1), \beta^{-1}(c_1)\right)(c_{n-1}, c_n). \tag{3.14}$$

□

Chapter 4

The 2×3 Game

The "2×3 game" is a simple puzzle with six tiles in a grid of numbers and a set of allowed moves which permute them. The goal of the game is to sort the grid into a "solved" state from some scrambled state.

$$\boxed{\begin{array}{ccc} \mathbf{5} & \mathbf{4} & \mathbf{2} \\ \mathbf{3} & \mathbf{6} & \mathbf{1} \end{array}} \overset{\text{Move}_1}{\longmapsto} \cdots \overset{\text{Move}_n}{\longmapsto} \boxed{\begin{array}{ccc} \mathbf{1} & \mathbf{2} & \mathbf{3} \\ \mathbf{4} & \mathbf{5} & \mathbf{6} \end{array}}$$

Formally, the game can and will herein henceforth be modelled as follows. Let $G := (\langle \mathfrak{M} \rangle, \circ)$ be a group compositionally generated by the set of moves $\mathfrak{M} := \{(123), (456), (14)\}$, named $U = (123)$ for *upper row*, $L = (456)$ for *lower row* and $S = (14)$ for *side flip*. A *problem* is a permutation $p \in G$, and a *solution* thereto is a vector $\mathbf{s} \in \mathfrak{M}^*$ such that

$$\bigcirc_{i=n}^{1} \mathbf{s}_i = \mathbf{s}_n \circ \mathbf{s}_{n-1} \circ \cdots \circ \mathbf{s}_1 = p$$

where n denotes the length of $\mathbf{s}$. Note that $\mathbf{s}$ may include the implicitly given inverses of the group generators. Hereafter, $\bigcirc_{i=n}^{1} \mathbf{s}_i$ will be written as $\acute{\mathbf{s}}$.

The size of the group $|G|$ is 720, which shows that $G = S_6$, since $|S_6| = 720$. This fact can also be proven in the following way. Each transposition beteen the upper and lower row can be found with the nested conjugate

$$\begin{aligned} T_1(x, y) &= \left(L^{-y+3+1}\right)^{-1} \left(\left(U^{-x+1}\right)^{-1} (S) \left(U^{-x+1}\right)\right) \left(L^{-y+3+1}\right) \\ &= L^{y-4} U^{x-1} S U^{-x+1} L^{-y+4} \\ &= L^{y-1} U^{x-1} S U^{-x+1} L^{-y+1}, \end{aligned}$$

where $x \in \{1,2,3\}$ and $y \in \{4,5,6\}$. As an example, let $p = (26)$. Then

$$\begin{aligned}
T_1(2,6) &= L^{6-1}U^{2-1}SU^{-2+1}L^{-6+1} \\
&= L^5U^1SU^{-1}L^{-5} \\
&= (456)^5(123)^1(14)(123)^{-1}(456)^{-5} \\
&= (465)(123)(14)(132)(456) \\
&= (465)(24)(456) \\
&= (2,6).
\end{aligned}$$

This gives – after some reductions – the solution $\mathbf{s} = \left(L, U^{-1}, S, U, L^{-1}\right)$ visualized below.

$$\boxed{\begin{array}{ccc} \mathbf{1} & \mathbf{6} & \mathbf{3} \\ \mathbf{4} & \mathbf{5} & \mathbf{2} \end{array}} \overset{L}{\longmapsto} \boxed{\begin{array}{ccc} \mathbf{1} & \mathbf{6} & \mathbf{3} \\ \mathbf{2} & \mathbf{4} & \mathbf{5} \end{array}} \overset{U^{-1}}{\longmapsto} \boxed{\begin{array}{ccc} \mathbf{6} & \mathbf{3} & \mathbf{1} \\ \mathbf{2} & \mathbf{4} & \mathbf{5} \end{array}}$$

$$\overset{S}{\longmapsto} \boxed{\begin{array}{ccc} \mathbf{2} & \mathbf{3} & \mathbf{1} \\ \mathbf{6} & \mathbf{4} & \mathbf{5} \end{array}} \overset{U}{\longmapsto} \boxed{\begin{array}{ccc} \mathbf{1} & \mathbf{2} & \mathbf{3} \\ \mathbf{6} & \mathbf{4} & \mathbf{5} \end{array}} \overset{L^{-1}}{\longmapsto} \boxed{\begin{array}{ccc} \mathbf{1} & \mathbf{2} & \mathbf{3} \\ \mathbf{4} & \mathbf{5} & \mathbf{6} \end{array}}$$

Each transposition beteen elements on the same row can be found with the nested conjugate

$$\begin{aligned}
T_2(x,y,R) &= \Big((R^{-x+1})^{-1}(S)(R^{-x+1})\Big)^{-1}\Big((R^{-y+1})^{-1}(S)(R^{-y+1})\Big)\Big((R^{-x+1})^{-1}(S)(R^{-x+1})\Big) \\
&= (R^{x-1}SR^{-x+1})^{-1}(R^{y-1}SR^{-y+1})(R^{x-1}SR^{-x+1}) \\
&= (R^{-(-x+1)}S^{-1}R^{-(x-1)})(R^{y-1}SR^{-y+1})(R^{x-1}SR^{-x+1}) \\
&= (R^{x-1}SR^{-x+1})(R^{y-1}SR^{-y+1})(R^{x-1}SR^{-x+1}) \\
&= R^{x-1}SR^{-x+1}R^{y-1}SR^{-y+1}R^{x-1}SR^{-x+1} \\
&= R^{x-1}SR^{-x+y}SR^{x-y}SR^{-x+1}
\end{aligned}$$

where $x, y \in R$ and $R \in \mathfrak{M} \setminus \{S\}$. As an example, let $p = (23)$. Since 2 and 3 belong to the upper row, R is set to U:

$$\begin{aligned}
T_2(2,3,U) &= U^{2-1}SU^{-2+3}SU^{2-3}SU^{-2+1} \\
&= U^1SU^1SU^{-1}SU^{-1} \\
&= (123)(14)(123)(14)(123)^{-1}(14)(123)^{-1} \\
&= (123)(14)(123)(14)(132)(14)(132) \\
&= (123)(14)(24)(14)(132) \\
&= (123)(12)(132) \\
&= (23).
\end{aligned}$$

This gives the solution $\mathbf{s} = \left(U^{-1}, S, U^{-1}, S, U, S, U\right)$ visualized below.

$$\boxed{\begin{matrix}\mathbf{1}&\mathbf{3}&\mathbf{2}\\\mathbf{4}&\mathbf{5}&\mathbf{6}\end{matrix}} \xmapsto{U^{-1}} \boxed{\begin{matrix}\mathbf{3}&\mathbf{2}&\mathbf{1}\\\mathbf{4}&\mathbf{5}&\mathbf{6}\end{matrix}} \xmapsto{S} \boxed{\begin{matrix}\mathbf{4}&\mathbf{2}&\mathbf{1}\\\mathbf{3}&\mathbf{5}&\mathbf{6}\end{matrix}} \xmapsto{U^{-1}} \boxed{\begin{matrix}\mathbf{2}&\mathbf{1}&\mathbf{4}\\\mathbf{3}&\mathbf{5}&\mathbf{6}\end{matrix}}$$

$$\xmapsto{S} \boxed{\begin{matrix}\mathbf{3}&\mathbf{1}&\mathbf{4}\\\mathbf{2}&\mathbf{5}&\mathbf{6}\end{matrix}} \xmapsto{U} \boxed{\begin{matrix}\mathbf{4}&\mathbf{3}&\mathbf{1}\\\mathbf{2}&\mathbf{5}&\mathbf{6}\end{matrix}} \xmapsto{S} \boxed{\begin{matrix}\mathbf{2}&\mathbf{3}&\mathbf{1}\\\mathbf{4}&\mathbf{5}&\mathbf{6}\end{matrix}} \xmapsto{U} \boxed{\begin{matrix}\mathbf{1}&\mathbf{2}&\mathbf{3}\\\mathbf{4}&\mathbf{5}&\mathbf{6}\end{matrix}}$$

Since a transposition of any two elements herein can be composed using the generators in $\mathfrak{M}$ – in other words, since the group G includes every transposition of the elements 1 through 6 – it follows that the group must include every permutation on the six elements, and thereby equal S_6.

Chapter 5

New

Since G is the symmetric group S_6, there must exist a map $T_1(x, y) = (x, y)$, where $x \in \{1, 2, 3\}$ and $y \in \{4, 5, 6\}$.

From Corrolary 1, we know that since $\zeta_1(x, y)$ is a transposition, then $\zeta_1 = \searrow^{\alpha_1}_{\beta_1}$ where $\beta_1 = (x, c_1)$ and $\alpha_1(y) = c_1$. Similarly, from $\left|\underline{\beta_1}\right| = 2$ follows that $\beta_1 = \searrow^{\alpha_2}_{\beta_2}$ where $\beta_2 = (c_1, c_2)$ and $\alpha_2(x) = c_2$. Now, from $\mathfrak{M}$ we can find permutations matching these properties

$$T_1(x, y) = \searrow^{L^{-y+3+1}}_{\searrow^{U^{-x+1}}_{S}}$$

$$\begin{aligned} T_2(x, y, R) &= \left((R^{-x+1})^{-1}(S)(R^{-x+1})\right)^{-1}\left((R^{-y+1})^{-1}(S)(R^{-y+1})\right)\left((R^{-x+1})^{-1}(S)(R^{-x+1})\right) \\ &= R^{x-1}SR^{-x+y}SR^{x-y}SR^{-x+1} \end{aligned}$$

Since G is the symmetric group S_6, there must be a map $\zeta : \{1, \cdots, 6\}^2 \to \mathcal{P}(\mathfrak{M}^{<\omega})$, where $\mathfrak{M}^{<\omega} = \bigcup_{n \in \mathbb{N}} \mathfrak{M}^n$ and $\mathbf{s} \in \zeta(x, y) \implies \acute{\mathbf{s}} = (x, y)$. In other words: there must exist a set of solutions to each transposition in S_6.

Since G is the symmetric group S_6, there must be an infinite[1] set of maps $\{1, \cdots, 6\}^2 \to \mathfrak{M}^{<\omega}$, where $\mathfrak{M}^{<\omega} = \bigcup_{n \in \mathbb{N}} \mathfrak{M}^n$ and $\mathbf{s} \in \zeta(x, y) \implies \acute{\mathbf{s}} = (x, y)$ where ζ is such a map.

::::::::::There is a set of solutions to T_1, and α_1 and β_1 and α_2 and β_2. the solution is the intersection thereof. At each level there is a uniqe map (for that level) to a set of solutions[2]. It's recursive!

—

The lemmas and corollaries in chapter 3 – and indeed, any other observation, general or particualar (about permutations or a specific game) – together construct a formal language.

Example: π where $|\underline{\pi}| = 2$ is a *pattern*, which can be substituted by another pattern: a conjugation $\searrow^{\alpha_1}_{\beta_1}$ where $\beta_1 = (x, c_1)$ and $\alpha_1(y) = c_1$ (as in the example above). The permutaion $\beta_1 = (x, c_1)$ is, then, also a pattern to be substituted and so on until you have a permutation constructed from available moves ($\mathfrak{M}$?). You don't even need to evaluate them all if you know, for example, that ζ_1/T_1 exists.

All the lemmas and corollaries are in fact rewritings (with rewriting rules – in this case, formal math) ...

A *pattern* is a set p. If a set, vector, element (or other data) Ω maches the pattern p, then $\Omega \in p$. The set $\{p \mid \Omega \in p\}$ is the set of patterns to which some data Ω belongs.

Example

Let $a \ggg b = [a_1, \cdots, a_n, b_1, \cdots, b_m]$, where $a = [a_1, \cdots, a_n]$ and $b = [b_1, \cdots, b_m]$.

$$\begin{array}{lcl} \mathfrak{E} & := & \mathfrak{T} \cup \{\mathfrak{e}_1 \ggg \mathfrak{o} \ggg \mathfrak{e}_2 \mid \mathfrak{o} \in \{[\text{“+”}], [\text{“−”}]\} \wedge \mathfrak{e}_1, \mathfrak{e}_2 \in \mathfrak{E}\} \\ \mathfrak{T} & := & \mathfrak{F} \cup \{\mathfrak{t}_1 \ggg \mathfrak{o} \ggg \mathfrak{t}_2 \mid \mathfrak{o} \in \{[\text{“×”}], [\text{“÷”}]\} \wedge \mathfrak{t}_1, \mathfrak{t}_2 \in \mathfrak{T}\} \\ \mathfrak{F} & := & \mathfrak{N} \cup \{[\text{“(”}] \ggg \mathfrak{e} \ggg [\text{“)”}] \mid \mathfrak{e} \in \mathfrak{E}\} \\ \mathfrak{N} & := & \mathcal{N} \cup \mathcal{N}^* \\ \mathcal{N} & := & \{\text{“0”}, \cdots, \text{“9”}\} \end{array}$$

$$\sigma' = \rotatebox[origin=c]{180}{ζ}(\Phi, \sigma)$$

$$\begin{cases} 1 & : x < 4 \\ 0 & : x > 6 \end{cases}$$

[1]It is easy to see that this set is infinite, since one can append any solution with a permutation and its inverse to get a new solution. Take the solution $\mathbf{s}_2 = (\alpha, \beta, \gamma)$ as an example, then $\acute{\mathbf{s}}_1 = \acute{\mathbf{s}}_2$ where $\mathbf{s}_2 = (\alpha, \beta, \gamma, \alpha, \alpha^{-1})$.

[2]These are *patterns* with rewriting rules!

Chapter 6

Going Further

Definition 14 (Grimoire)**.** A *grimoire* Φ of a semigroup G, a set of *spells* $\mathfrak{S}_\Phi \subseteq G$ and a set of *movements* $\mathfrak{M} \subseteq \mathfrak{S}_\Phi$ is a map in $\mathfrak{S}_\Phi \to \mathcal{P}\left(\mathfrak{S}_\Phi^{<\omega}\right)$ such that $\mathfrak{s}_1 * \cdots * \mathfrak{s}_n = \mathfrak{s}$ – where $*$ is the binary operator belonging to G – for each $[\mathfrak{s}_1, \cdots, \mathfrak{s}_n] \in \Phi(\mathfrak{s})$ and $\mathfrak{s} \in \mathfrak{S}_\Phi \setminus \mathfrak{A}$.

Definition 15 (Exploration Function)**.** Let a map $\triangle : \boxed{\Phi} \times G^{<\omega} \to \boxed{\Phi}$ – where $\boxed{\Phi}$ is the set of all grimoires of a subgroup G – be defined for some grimoire Φ by $\forall \mathfrak{s} \in \mathfrak{S}_\Phi \setminus \{\mathfrak{s}_1 * \cdots * \mathfrak{s}_n\}$ $(\Phi'(\mathfrak{s}) = \Phi(\mathfrak{s}))$ and $\Phi'(\mathfrak{s}_1 * \cdots * \mathfrak{s}_n) = \Phi(\mathfrak{s}_1 * \cdots * \mathfrak{s}_n) \cup \{[\mathfrak{s}_1, \cdots, \mathfrak{s}_n]\}$ if $\mathfrak{s}_1 * \cdots * \mathfrak{s}_n \in \mathfrak{S}_\Phi$ and $\Phi'(\mathfrak{s}_1 * \cdots * \mathfrak{s}_n) = \{[\mathfrak{s}_1, \cdots, \mathfrak{s}_n]\}$ if not – where $\Phi' = \triangle(\Phi, [\mathfrak{s}_1, \cdots, \mathfrak{s}_n])$ and $\mathfrak{s}_1, \cdots, \mathfrak{s}_n \in \mathfrak{S}_\Phi$. (Note that since $\mathfrak{S}_\Phi = \mathrm{dom}(\Phi)$, then $\mathfrak{S}_{\Phi'} = \mathfrak{S}_\Phi \cup \{\mathfrak{s}_1 * \cdots * \mathfrak{s}_n\}$ if $\Phi' = \triangle(\Phi, [\mathfrak{s}_1, \cdots, \mathfrak{s}_n])$.) An *exploration function* $\mathscr{E}$ in G is a map in $\boxed{\Phi} \times \boxed{C}^{<\omega} \to \boxed{\Phi}$ defined by

$$\mathscr{E}(\Phi, [C_1, \cdots, C_n]) := \bigodot_{i=1}^{n} \triangle\left(\Phi, C_i\left(\mathfrak{S}_\Phi\right)\right),$$

where $\boxed{C} \subset \mathcal{P}(G) \to G^{<\omega}$ and $C(S) \in S^{<\omega}$ for each $C \in \boxed{C}$ and $S \in \mathcal{P}(G)$.

Definition 16 (Expansion Function)**.** An *expansion function* $\wr$ for a grimoire Φ of a semigroup G is a map in $\boxed{\Phi} \times \mathfrak{S}_\Phi \times \boxed{\Pi} \to \mathfrak{M}^{<\omega}$ – where $\boxed{\Phi}$ is the set of all grimoires of G – defined by

$$\wr(\Phi, \mathfrak{s}, \Pi) := \begin{cases} \mathfrak{s} & : \wr_{\text{step}}(\Phi, \mathfrak{s}, \Pi) = \mathfrak{s} \\ \wr\left(\Phi, \wr_{\text{step}}(\Phi, \mathfrak{s}, \Pi), \Pi\right) & : \text{otherwise} \end{cases}$$

where $\wr_{\text{step}} : \boxed{\Phi} \times \mathfrak{S}_\Phi \times \boxed{\Pi} \to \mathfrak{S}_\Phi^{<\omega}$ is defined by

$$\wr_{\text{step}}(\Phi, \mathfrak{s}, \Pi) := \mathop{\gg}_{i=1}^{|\mathfrak{s}|} \left(\begin{cases} \mathfrak{s}_i & : \mathfrak{s}_i \in \mathfrak{M} \\ \Pi\left(\Phi\left(\mathfrak{s}_i\right)\right) & : \text{otherwise} \end{cases} \right),$$

where $\boxed{\Pi} \subset \mathcal{P}\left(G^{<\omega}\right) \to G^{<\omega}$ is a set of maps wherein $\Pi(S) \in S$ for each $\Pi \in \boxed{\Pi}$ and $S \in \mathcal{P}\left(G^{<\omega}\right)$ hold.